Coordinates in 2D - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Write down the equation of the horizontal line through $(3, 7)$.

Every point has $y = 7$

Answer: y = 7

Question 2 *Practice question written for this worksheet (not from a past paper)*

Write down the equation of the vertical line through $(-2, 5)$.

Every point has $x = -2$

Answer: x = -2

Question 3 *Practice question written for this worksheet (not from a past paper)*

Work out the length of AB where $A = (-3, 2)$ and $B = (5, -4)$.

Differences: 8 and 6

$\sqrt{64 + 36}$

Answer: 10

Question 4 *Practice question written for this worksheet (not from a past paper)*

Find where the line $y = 3x - 6$ crosses the x-axis and the y-axis.

$y = 0$: $3x = 6$, $x = 2$

$x = 0$: $y = -6$

Answer: (2, 0) and (0, -6)

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Question 5

Practice question written for this worksheet (not from a past paper)

Find where the line $2x + 5y = 20$ crosses the axes.

$y = 0$: $2x = 20$, $x = 10$

$x = 0$: $5y = 20$, $y = 4$

Answer: (10, 0) and (0, 4)

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 1

Values in quotation marks must come from a correct method previously seen unless clearly stated otherwise.

Q	Working	Answer	Mark	Notes
1	$(x=)\frac{3+9}{2}\left(=\frac{12}{2}=6\right)$ oe or $(y=)\frac{4+16}{2}\left(=\frac{20}{2}=10\right)$ oe or $(x=)$ 6 or $(y=)$ 10 or (10, 6)		2	M1 for a **correct** method to work out the midpoint of x or the midpoint of y or one correct value or (10, 6)
	Correct answer scores full marks (unless from obvious incorrect working)	(6, 10)		A1
				Total 2 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 4

4	$\frac{-4+1}{2}$ **or** $\frac{9+5}{2}$			M1	or for $(-1.5, y)$ or $(x, 7)$ or $(7, -1.5)$
		$(-1.5, 7)$	2	A1	oe
					Total 2 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 6

6	e.g. $a = (-3 + 47) \div 2$ $(= 22)$ **or** $\frac{11+b}{2} = -19$ $(b = -38 - 11 = -49)$ **or** method to add 25 to -3 **or** method to subtract 25 from 47 **or** method to subtract 30 from -19 **or** method to subtract 60 from 11		2	M1	for a correct method to find either coordinate or one coordinate correct. Look for correct method on their diagram, if used.
		$a = 22$, $b = -49$		A1	both correct
					Total 2 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 6

6	$0.5 \times (1 + 6)$ or $0.5 \times (4 + 9)$		2	M1	for a correct method to find one coordinate or for 1 coordinate given correctly or for (6.5, 3.5)
		(3.5 , 6.5)		A1	
					Total 2 marks

Coordinates in 2D - Mark scheme

Question 10 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 6*

Question	Working	Answer	Mark	Notes
6	$[k=]\ \frac{6+17}{2}$ **or** $[k=]6+\frac{17-6}{2}$ oe **or** $[j=]\ 4+2(15-4)$ **or** $[j=]\ 15+(15-4)$ **or** $\frac{4+j}{2}=15$ oe		3	M1
	Correct answers score full marks (unless from obvious incorrect working) *1 correct answer will score M1A1 and both will score M1A1A1*	26		A1
		11.5		A1 oe eg $\frac{23}{2}$ both answers the wrong way round scores M1A1 unless the correct answers are clearly labelled in working space
				Total 3 marks

Question 11 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2017 · Question 6*

6 (a)		6	1	B1	
(b)	$(y=)\frac{3e+7e}{2}$ or $(y=)\frac{10e}{2}$ oe or $(y=)3e+\frac{7e-3e}{2}$ or $(y=)3e+2e$ oe or $(y=)7e-\frac{7e-3e}{2}$ or $(y=)7e-2e$ oe		2	M1	For an unsimplified expression equivalent to $5e$
		$5e$		A1	cao
					Total 3 marks

Question 12 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 10*

10 (a)	$(18-3)^2+(7--1)^2$ oe or $15^2+8^2\ (=289)$ oe		3	M1	
	$\sqrt{(18-3)^2+(7--1)^2}$ $(=\sqrt{"289"})$			M1	
		17		A1	
(b)	13 + 6 > "17"	correct reason	1	A1ft dep M1	**Acceptable examples** "They overlap by 2cm" "The distance between the centres is less than the sum of the radii" "17 is less than the distance than the total of the radii" "19 is bigger than the distance between the centres" **Not acceptable examples** "19 is greater than the distance between the circles" oe "The circumference of each circle overlaps"
					Total 4 marks

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2023 · Question 4*

Q	Working	Answer	Mark	Notes
(a)		(1, 6)	1	B1
(b)		(−3, −1)	1	B1
(c)		Cross at (1, 2)	1	B1
(d)		Cross at (5, −1)	1	B1
(e)		1	1	B1 cao
				Total 5 marks

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Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 9

Q	Working	Answer	Mark	Notes
(a)		(–4, 2)	1	B1 cao
(b)		(–1, 3)	2	B2 (B1 for $(-1, y)$ **or** $(x, 3)$ **or** $(3, -1)$ **or** the midpoint of AB unambiguously shown on diagram)
(c)		(–3, –1)	2	B2 (B1 for $(-3, y)$ **or** $(x, -1)$ **or** $(-1, -3)$ **or** D marked correctly on the diagram)
				Total 5 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 22

22	(gradient AB =) $\frac{10--5}{p--1}$ $\left(=\frac{10+5}{p+1}=\frac{15}{p+1}\right)$ oe or (gradient BC =) $\frac{q--5}{8--1}$ $\left(=\frac{q+5}{8+1}=\frac{q+5}{9}\right)$ oe or (gradient AC =) $\frac{10-q}{p-8}$ oe **or** $\sqrt{(p--1)^2+(10--5)^2}$ or $(p--1)^2+(10--5)^2$ or $\sqrt{(8--1)^2+(q--5)^2}$ or $(8--1)^2+(q--5)^2$ or $\sqrt{(8-p)^2+(q-10)^2}$ or $(8-p)^2+(q-10)^2$ oe		5	M1 for finding the gradient of AB or BC or AC This may be seen embedded in $m_1 \times m_2 = -1$ **or** for finding the length of AB or BC or AC (or AB^2 etc)
	• $\frac{15}{p+1}\times\frac{q+5}{9}=-1$ or $\frac{15}{p+1}=-\frac{9}{q+5}$ or $9p+15q=-84$ oe • $\frac{10-q}{p-8}=-\frac{6}{7}$ or $6p-7q=-22$ oe • $(p--1)^2+(10--5)^2+(8--1)^2+(q--5)^2=(8-p)^2+(q-10)^2$ or $18p+30q=-168$ **Alternative for the second point** • $\frac{6}{7}p+10=-8\times-\frac{6}{7}+q$ oe			M2 for two out of the three of: • using $m_1 \times m_2 = -1$ • using the gradient of AC to form an equation. • using Pythagoras theorem If not M2, then M1 for one of the equations. **Alternative for the second point** obtaining this equation by using $y = mx + c$ with coordinates of A and C separately, and then eliminating c)
	Elimination E.g. $54p + 90q = -504$ $54p - 63q = -198$ With subtraction **or** $153q = -306$ **or** $63p + 105q = -588$ $90p - 105q = -330$ With the operation of addition **or** $153p = -918$ Substitution E.g. $6\left(\frac{-84-15q}{9}\right)=-22$ or $6p-7\left(\frac{-84-9p}{15}\right)=-22$ or $9\left(\frac{-22+7q}{6}\right)+15q=-84$ or $9p+15\left(\frac{6p+22}{7}\right)=-84$			M1 (dep M3) for correct method to eliminate one variable – multiplying one or both equations so the coefficient of x or y is the same in both, with the correct operation to eliminate one variable (condone one arithmetic error) **or** isolating p or q in one equation and substituting into the other (condone one arithmetic error).
		$p = -6$ and $q = -2$		A1 for $p = -6$ and $q = -2$ Must be clearly identified
				Total 5 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2022 · Question 7

Q	Working	Answer	Mark	Notes
(a)		(−1, 3)	1	B1
(b)		(5, 1)	2	B1 for $x = 5$ B1 for $y = 1$
(c)	$\frac{1}{2}\times 6\times 4$ oe		2	M1 for a correct method
		12		A1
(d)		D indicated at (−1, −1)	1	B1 label not required if coordinate clearly indicated
				Total 5 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 7

7 (a)		(4.5 , 3)	2	B1 B1
(b)	Identifies 2 & 7 as sides "2"^{2} + "7"^{2} √("2"^{2} + "7"^{2})	7.28	4	B1 M1 "2" & "7" must be identified as sides M1 dep A1 awrt 7.28
				Total 6 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 25

25	(gradient of AB =) "$-\frac{1}{2}$" or "2" $m = -1$		6	M1 for the use of $m_1 \times m_2 = -1$ or for "$-\frac{1}{2}$" embedded in a linear equation eg $y =$ "$-\frac{1}{2}$"$x + c$
	(gradient of AB =) $\frac{k-7}{6-j}$ oe **or** (midpoint of AB =) $\left(\frac{j+6}{2}, \frac{k+7}{2}\right)$ oe			M1 for a correct expression for the gradient which may be seen in an equation **or** for a correct expression for the midpoint which may be seen in an equation.
	$\frac{k-7}{6-j} = -\frac{1}{2}$ oe or $2k - j = 8$ oe **or** $\left(\frac{k+7}{2}\right) - 2\left(\frac{j+6}{2}\right) = 7$ oe or $k - 2j = 19$ oe			M1 for setting up a **correct** equation for AB in terms of gradient **or** for setting up a **correct** equation for the line given and the midpoint
	$\frac{k-7}{6-j} = -\frac{1}{2}$ oe or $2k - j = 8$ oe **and** $\left(\frac{k+7}{2}\right) - 2\left(\frac{j+6}{2}\right) = 7$ oe or $k - 2j = 19$ oe			A1 for 2 **correct** equations
	$k = -1$ **and** $j = -10$			A1 for a correct value of k **and** a correct value of j
	Working required	(−2, 3)		A1 dep on previous M1
				Total 6 marks

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Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 25

25	eg $\left(\frac{-4+2}{2}, \frac{6+3}{2}\right)$ **or** (−1, 4.5) oe		6	M1	for method to find the midpoint of *AB*
	eg $\frac{6-3}{-4-2}\left(=\frac{3}{-6}\right)$ oe **or** $-\frac{1}{2}$ oe **or** −0.5			M1	for method to find the gradient of *AB*
	eg $m \times "-0.5" = -1$ oe **or** $m = 2$			M1	for use of $m_1 m_2 = -1$ to find the gradient of the line of symmetry
	eg $y-8="-0.5"(x-(-1))$ **or** 8 = "−0.5" × −1 + *c* **or** $\frac{y-8}{x-(-1)}="-0.5"$ **or** $y-4.5="2"(x-(-1))$ **or** 4.5 = "2" × −1 + *c* **or** $\frac{y-4.5}{x-(-1)}="2"$			M1	for method to find an equation for *CD* **or** the line of symmetry
	eg $2x + 6.5 = -0.5x + 7.5$ **or** $\frac{y-6.5}{2}=\frac{y-7.5}{-0.5}$			M1	for a correct linear equation to find the *x* or *y* coordinate of *E*
	Correct answer scores full marks (unless from obvious incorrect working)	(0.4, 7.3)		A1	oe
					Total 6 marks

ALTERNATIVE (using the length of *CD*):

25	eg $\frac{6-3}{-4-2}\left(=\frac{3}{-6}\right)$ oe **or** $-\frac{1}{2}$ oe **or** −0.5		6	M1	for method to find the gradient of *AB*
	eg $y-8="-0.5"(x+1)$ **or** 8 = "−0.5" × −1 + *c* **or** $\frac{y-8}{x-(-1)}="-0.5"$			M1	for method to find an equation for *CD*
	eg $\sqrt{(-1-(-4))^2+(8-6)^2}\ (=\sqrt{13})$			M1	for method to find the length of *AD* or AD^2
	eg $\sqrt{(x-2)^2+(7.5-0.5x-3)^2}="\sqrt{13}"$ **or** $\sqrt{(15-2y-2)^2+(y-3)^2}="\sqrt{13}"$			M1	for setting up an equation for the *x* or *y* coordinate of *C*
	(1.8, 6.6) oe			M1	for the correct coordinates for *C*
	Correct answer scores full marks (unless from obvious incorrect working)	(0.4, 7.3)		A1	oe
					Total 6 marks

CHALLENGE QUESTIONS

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 18

18	(a)			4	B1	$b = 14$
		(Gradient AB =) $\frac{12}{5}$ oe or eg $\frac{10--2}{1--4}$ oe			M1	For the gradient of *AB*
		(Gradient BC =) $-\frac{5}{12}$ oe			M1	Ft correct use of $m_1 \times m_2 = -1$ for *their* gradient of *AB* or $a = 2.5$ or $c = -9.5$
			$a = 2.5$, $c = -9.5$		A1	for $a = 2.5$ and $c = -9.5$
	(b)	$(AB =) \sqrt{(1--4)^2+(10--2)^2}$ $(=\sqrt{5^2+12^2}\ (=13))$		3	M1	
		$(BC =) \sqrt{(19-1)^2+(10-2.5)^2}$ $(=\sqrt{18^2+7.5^2}\ (=19.5))$ **or** $\sqrt{(19-1)^2+(10-\text{their } a)^2}$ **or** 1.5 × "13"			M1	ft their value of *a*
			65		A1	
						Total 7 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 22

Question	Working	Answer	Mark	Notes
	gradient of $JK = -0.5$ **or** $m \times 2 = -1$		6	M1 for finding the gradient of JK using $m_1 \times m_2 = -1$
	$\frac{k-15}{6-j} = -\frac{1}{2}$ **or** $2k - j = 24$ **or** $j = 2k - 24$ **or** $k = \frac{j+24}{2}$ oe			M1 for expressing the gradient of JK in terms of j and k or a correct equivalent equation

	$j = 2k - 24$ or $k = \frac{\quad}{2}$ oe			equivalent equation
	$(j-6)^2 + (k-15)^2 = 80$ oe **or** $\left(\frac{j+6}{2}, \frac{k+15}{2}\right)$ oe **or** $(j+4)^2 + 196 = 100 + (k-1)^2$ oe			M1 for finding equation of JK in terms of j and k **or** for finding the midpoint of M **or** for equating length HJ with length HK
	eg $3k^2 - 78k + 495 = 0$ oe **or** $5j^2 - 60j - 140 = 0$ oe **or** $5k^2 - 150k + 1045 = 0$ oe **or** $3j^2 - 12j - 36 = 0$ **oe** **or** gradient HM: eg $\frac{\frac{k+15}{2} - 1}{\frac{j+6}{2} + 4} = 2$ **or** $k = 2j + 15$ **or** $j = \frac{k-15}{2}$ oe			M1 (dep on M3) writing a correct quadratic expression in the form $ax^2 + bx + c$ $(= 0)$ (allow $ax^2 + bx = c$) **or** A correct equation for the gradient of HM in terms of j and k or a correct equivalent equation

	eg $(k-15)(k-11)(=0)$ **or** $\frac{78 \pm \sqrt{(-78)^2 - 4\times3\times495}}{2\times3}$ **or** $(k-13)^2 - 169 + 165 (=0)$	eg $(j-6)(j+2)(=0)$ **or** $\frac{12 \pm \sqrt{(-12)^2 - 4\times3\times-36}}{2\times3}$ **or** $(j-2)^2 - 4 - 12 (=0)$			M1 (dep on M3) for a complete method to solve their 3-term quadratic equation (allow one sign error in the use of the quadratic formula) **or** a correct method to eliminate either j or k eg $2k - 24 = \frac{k-15}{2}$ oe **or** $\frac{j+24}{2} = 2j + 15$ oe
		$j = -2, k = 11$			A1
					Total 6 marks

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Question	Working	Answer	Mark	Notes
	$\left(\frac{j+6}{2}, \frac{k+15}{2}\right)$ oe		6	M1 for finding the midpoint of M
ALT	$\frac{\frac{k+15}{2}-1}{\frac{j+6}{2}+4} = 2$ **or** $k-2j=15$ **or** $k=2j+15$ **or** $j=\frac{k-15}{2}$ oe			M1 for expressing the gradient of JK in terms of j and k or a correct equivalent equation
	$(j-6)^2+(k-15)^2=80$ oe **or** $(j+4)^2+196=100+(k-1)^2$ oe			M1 for finding the length of JK in terms of j and k **or** for equating length HJ with length HK
	E.g. $5j^2-12j-44=0$ **or** $3j^2+48j+84=0$ oe E.g. $5k^2-174k+1309=0$ **or** $3k^2+6k-429=0$ oe			M1 (dep on M3) writing the correct quadratic expression in form $ax^2+bx+c\ (=0)$ allow $ax^2+bx=c$
	E.g. $(5j-22)(j+2)(=0)$ **or** $\frac{12\pm\sqrt{(-12)^2-4\times5\times-44}}{2\times5}$ **or** $(j+8)^2-64+28(=0)$ E.g. $(5k-119)(k-11)(=0)$ **or** $\frac{174\pm\sqrt{(-174)^2-4\times5\times1309}}{2\times5}$ **or** $(k+1)^2-1-143(=0)$			M1 (dep on M3) for a complete method to solve their 3-term quadratic equation (allow one sign error in the use of the quadratic formula)
		$j=-2,\ k=11$		A1
				Total 6 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 23*

Q	Working	Answer	Mark	Notes
	$y=\frac{3}{5}x\left(+\frac{6}{5}\right)$ or $y=0.6x(+1.2)$ or (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k=-2$			B1 for $k=-2$
	$\left(\frac{"-2"+8}{2}, \frac{1+7}{2}\right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k<0$)
	$"\frac{3}{5}"m=-1$ **or** $(m=)"-\frac{5}{3}"$			M1ft their gradient for use of $m_1\times m_2=-1$ Allow $-\frac{5}{3}=-1.67$ or better
	$"4"="-\frac{5}{3}"\times"3"+c$ or $c=9$ or $y-"4"="-\frac{5}{3}"(x-"3")$			M1 dep on M3
	Working required	$5x+3y=27$		A1 allow equation in any form where p, q and r are integers
				Total 6 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 24

24	$\frac{-5-10}{4--2}(=-\frac{5}{2})$		6	M1	A correct method to find the gradient of AC
	$y-10=-\frac{5}{2}(x+2)$ oe eg $y=-\frac{5}{2}x+5$ or $y--5=-\frac{5}{2}(x-4)$ oe or $5x+2y=10$ oe			M1	ft (if M1 scored) correct equation of AC
	$y-4=\frac{2}{5}(x--\frac{27}{5})$ oe or $4=\frac{2}{5}\left(-\frac{27}{5}\right)+c\left(y=\frac{2}{5}x+6.16\right)$ $\frac{4-y}{-\frac{27}{5}-x}=\frac{2}{5}$ oe or $5y-2x=\frac{154}{5}$ oe			M1	ft (if first M1 scored) equation of BD or correct equation using gradient of BD
	solves $-\frac{5}{2}x+5=\frac{2}{5}x+6.16$ oe eg $\begin{matrix}10x+4y=20\\-10x+25y=154\end{matrix}$ oe, with operation of addition **or** $\begin{matrix}25x+10y=50\\-4x+10y=61.6\end{matrix}$ oe, with operation of subtraction **or** $x=\frac{5}{2}y-\frac{154}{10}$ oe or $y=\frac{2}{5}x+\frac{154}{25}$ oe substituted in other equation			M1	Solve equation OR Solve simultaneously the correct equations of lines of AC and BD or correct equation from gradient or other correct equation. If elimination: same coefficient of x or y with suitable sign used to eliminate. If substitution: x or y substituted into other equation.
	Coordinates of intersection of AC and BD : $x=-\frac{2}{5}$, $y=6$			M1	oe value of x and y at intersection of AC and BD
	Correct answer scores full marks (unless from obvious incorrect working)	(4.6, 8)		A1	oe coordinates of D
	See next page for working with $AD = AB$, $CD = CB$ or gradients				**Total 6 marks**

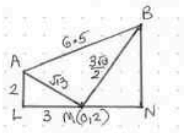
24	eg $(10-4)^2+\left(-2+\frac{27}{5}\right)^2$ $(=47.56)$ $(AB = 6.896\ldots)$ **or** eg $(-5-4)^2+(4+\frac{27}{5})^2(=169.36)$ $(CB = 13.013\ldots)$ **or** eg $\frac{-5-10}{4--2}$ **or** $\frac{4-y}{-\frac{27}{5}-x}$ oe		6	M1	A correct method to find AB or CB or AB or CB or a correct gradient expression for AC or DB
	eg $(y-10)^2+(x+2)^2=(10-4)^2+\left(-2+\frac{27}{5}\right)^2$ **or** eg $(y+5)^2+(x-4)^2=(-5-4)^2+(4+\frac{27}{5})^2$ **or** $\frac{-5-10}{4--2}\times\frac{4-y}{-\frac{27}{5}-x}=-1$ oe eg $-60+15y=6x+32.4$			M1	Using D (x, y) form a correct equation $AD = AB$ **or** $CD = CB$ **or** gradients $AC \times DB = -1$ (Using D (x, y))
	eg $2x-5y=-30.8$ **or** $x=2.5y-15.4$ **or** $y=0.4x+6.16$ oe			M1	uses rearrangement or solving simultaneous equations to find a correct 3 term linear equation
	eg $(y-10)^2+(2.5y-15.4+2)^2=(10-4)^2+\left(-2+\frac{27}{5}\right)^2$ eg $(0.4x+6.16+5)^2+(x-4)^2=(-5-4)^2+(4+\frac{27}{5})^2$			M1	uses substitution to obtain a correct quadratic equation in one unknown
	$7.25y-87y+232=0$ oe or $1.16x+0.928x-28.8144=0$ oe			M1	for a 3 term quadratic that can be used to find the value of x or the value of y at D
		(4.6, 8)		A1	oe coordinates of D
					Total 6 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 22

22	(gradient AM =) $\frac{4-2}{-3-0}$ oe (= $-\frac{2}{3}$)		7	M1	A correct method to find gradient of AM
	$y=\frac{3}{2}x+2$ or eg $\frac{y-2}{x}=\frac{3}{2}$ oe			M1	For the correct equation of the line passing through BD or for a correct expression involving the x and y coordinates of point B or point D
	$(x--3)^2+(y-4)^2=6.5^2$ or $(x-0)^2+(y-2)^2=6.5^2-[(-3-0)^2+(4-2)^2]$ oe eg $x^2+(y-2)^2=29.25$			M1	A correct equation in x and y to find the coordinates of B and D
	eg $x^2+6x+9+y^2-8y+16-42.25=0$ oe or $x^2+y^2-4y+4-29.25=0$ oe			M1	Brackets expanded
	eg $x^2+6x+9+\left(\frac{3}{2}x+2\right)^2-8\left(\frac{3}{2}x+2\right)+16-42.25=0$ $\left(\frac{2y-4}{3}\right)^2+y^2-4y+4-29.25=0$ oe			M1	For a correct substitution into a correct equation to get an equation in either x only or y only
	eg $\frac{13}{4}x^2=\frac{117}{4}$ or $13y^2-52y-211.25=0$ oe			M1	A fully correct simplified equation in x or in y – all brackets expanded and like terms grouped.
		(3, 6.5) (−3, −2.5)		A1	correct coordinates SCB3 for one pair of correct coordinates or both x values correct or both y values correct
					Total 7 marks

See next page for alternative scheme

22 Alt 1	$(AM=)\sqrt{3^2+2^2}$ (= $\sqrt{13}$ = 3.605...) or $(AM^2=)3^2+2^2(=13)$		7	M1	Use of Pythagoras for point A to point M
	$(BM=)\sqrt{6.5^2-"\sqrt{13}"^2}$ (= $\sqrt{29.25}=\frac{3\sqrt{13}}{2}$ 5.4083...)			M1	A correct method to find the length of BM or DM
	$(SF=)\frac{\sqrt{29.25}}{\sqrt{13}}=\frac{3}{2}$ oe **or** $MN=x$, $BN=1.5x$ (*see diag*) or $(LAM=)\sin^{-1}\frac{3}{\sqrt{13}}(=56.3...)$ oe **or** $(LMA=)\cos^{-1}\frac{3}{\sqrt{13}}(=33.6...)$ **or**			M1	A correct method to find the SF of the enlargement of the sides AM to BM or angle LAM OR LMA 
	eg $\overline{MB}_x=\frac{3}{2}\times2$ or $\overline{MB}_y=\frac{3}{2}\times3$ or $\overline{MD}_x=-\frac{3}{2}\times2$ or $\overline{MD}_y=-\frac{3}{2}\times3$ oe or $x^2+(1.5x)^2=\sqrt{29.25}^2$ **or** $MN=\sqrt{29.25}\cos56.3...(=3)$ oe **or** $BN=\sqrt{29.25}\sin56.3...(=4.5)$ oe *turn over*			M1	A correct method to find the translation of at least one component of MB or MD (need not be written in vector form) OR correct Pythagoras statement using the SF to find x coordinates OR 1 correct trig statement to find translations from M
	$\overline{MB}_x=\frac{3}{2}\times2$ **and** $\overline{MB}_y=\frac{3}{2}\times3$ **or** $\overline{MD}_x=-\frac{3}{2}\times2$ **and** $\overline{MD}_y=-\frac{3}{2}\times3$ oe **or** $x^2+2.25x^2=29.25$ **or** $MN=\frac{3\sqrt{13}}{2}\cos56.309...(=3)$ **and** $BN=\frac{3\sqrt{13}}{2}\sin56.309...(=4.5)$ oe			M1	A correct method to find the translation of both components of MB or MD (need not be written in vector form) OR correct Pythagoras statement with no brackets using the SF to find x coordinates OR 2 correct trig statements to find translations from M
	eg (0, 2) is translated $\begin{pmatrix}3\\4.5\end{pmatrix}$ or $(0+3, 2+4.5)$ (= (3, 6.5)) or (0, 2) is translated $\begin{pmatrix}-3\\-4.5\end{pmatrix}$ or $(0-3, 2-4.5)$ (= (−3, −2.5)) oe **or** $3.25x^2=29.25$			M1	correct method to find the coordinates of B or D or one pair of correct coordinates or a correct method to find both x coordinates or both y coordinates OR a fully correct simplified equation in x all brackets expanded and like terms grouped.
		(3, 6.5) (−3, −2.5)		A1	correct coordinates SCB3 for one correct coordinate or both x values correct or both y values correct
					Total 7 marks